

# Basics of matrices

## Key Points

A matrix of order  $n \times m$  has  $n$  rows and  $m$  columns

The identity matrix  $I_n$  is an  $n \times n$  matrix with 1's on the leading diagonal and zeros elsewhere

The zero matrix  $O$  is a matrix full of zeroes

To add or subtract matrices do so element by element

To multiply a matrix by a scalar multiply all elements by the scalar

## Basic Skills

**Q1.** Write down the  $2 \times 2$  identity matrix

**Q2.** Write down the  $4 \times 3$  zero matrix

**Q3.** Compute the following

a)  $\begin{pmatrix} 3 & 4 \\ 0 & 2 \end{pmatrix} + \begin{pmatrix} 1 & 5 \\ 2 & 1 \end{pmatrix}$

b)  $\begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix} - \begin{pmatrix} 1 & 0 \\ 2 & 1 \end{pmatrix}$

c)  $3 \begin{pmatrix} 2 & -1 \\ 4 & 0 \end{pmatrix}$

**Q4.** What must be true about the order of two matrices  $A$  and  $B$  to be able to add or subtract them

## Competency Skills

**Q5.** What is the name for matrices that are of order  $n \times n$  for any integer  $n$

**Q6.** Explain what it means for two matrices to be equal

**Q7.** Compute the following

a)  $2 \begin{pmatrix} 5 & 2 \\ -1 & 4 \end{pmatrix} - 4 \begin{pmatrix} -2 & 10 \\ 1 & 0 \end{pmatrix}$  b)  $\begin{pmatrix} -1 & 2 & -5 \\ 2 & -1 & 0 \\ 0 & 1 & 4 \end{pmatrix} + 3 \begin{pmatrix} 0 & 3 & 5 \\ -2 & 1 & -2 \\ -4 & -3 & 6 \end{pmatrix}$  c)  $\begin{pmatrix} 2 & 5 & 3 \\ -1 & 6 & 0 \end{pmatrix} - 2 \begin{pmatrix} 5 & -4 & -1 \\ -2 & -1 & 6 \end{pmatrix}$

**Q8.** Find all values of  $x, y$  such that  $\begin{pmatrix} x^2 & 5 \\ 2 & y \end{pmatrix} - 3 \begin{pmatrix} x & -3 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 2x & 14 \\ -1 & 4 \end{pmatrix}$

**Q9.** Find all values of  $x, y$  such that  $\begin{pmatrix} y & 2x \\ 2 & 0 \end{pmatrix} - \begin{pmatrix} 1 & y \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} x & 1 \\ 0 & -2 \end{pmatrix}$

## Advanced Skills

**Q10.** Given that the trace  $tr(A) = \sum a_{ii}$  of a square matrix  $A$  is the sum of the leading diagonal

a) Find the trace of  $\begin{pmatrix} -1 & 0 & -5 \\ 2 & 3 & 2 \\ -4 & -3 & 6 \end{pmatrix}$

b) Prove or disprove  $tr(A + B) = tr(A) + tr(B)$  for any two matrices  $A, B$

c) Prove or disprove  $tr(kA) = ktr(A)$  for a scalar  $k$

**Q11.** Determine if matrix addition is commutative

**Q12.** A student comments that "all vectors are matrices". Comment on their statement

## Extension

**Q13.** Matrices can be defined to contain other element fields such as complex numbers. Determine how it is possible to represent a complex matrix as two real matrices

Hint: Consider the real part and imaginary part